

Total No. of Questions : 8]

SEAT No. :

PB2525

[6263]-423

[Total No. of Pages :2

B.E. (E & TC) (Honors in Robotics)
INDUSTRIAL ROBOTICS AND AUTOMATION
(2019 Pattern) (Semester-VII) (404181 HR)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume suitable data, wherever necessary.

Q1) a) Draw and explain single acting and double acting cylindrical actuators with proper operation. [6]

b) List different methods to control pneumatic control system. What is Karnaugh - Veitch mapping. [6]

c) List advantages and disadvantages of pneumatic system. [6]

OR

Q2) a) Draw different types of valves symbols and explain any one with the example. [6]

b) Enlist the basic components in pneumatic system and explain them in detail. [6]

c) What is pneumatic system? List the industrial applications of it and elaborate any one. [6]

Q3) a) Write the steps to design of parts for high speed feeding and orienting. State different feeding difficulties. [9]

b) Explain in detail how high speed automation insertion is possible in robotic assembly. State general rules for product design and automation. [8]

OR

Q4) a) How to perform the analysis of an assembly. Explain with one example. [9]

b) State design features of CNC systems. Explain drive system for CNC machine tools. [8]

P.T.O.

- Q5)** a) Write a case study on engine management system. [6]
b) State and explain different stages in designing a mechatronics system. [6]
c) Write a case study on Pick and Place robot. [6]

OR

- Q6)** a) How mechatronics system design is different than traditional system? [6]
b) Give the classification of a mechatronics system. [6]
c) Define Mechatronics. Explain different key elements of the Mechatronics system. [6]

- Q7)** a) With a suitable diagram and PLC ladder diagram explain how to control two cylinders? [9]

- b) With an example explain the calculations for power and velocity during extension and retraction. [8]

OR

- Q8)** a) Classify the pumps. Give characteristics and selection criteria for pumps. [9]

- b) Classify the control valves used in hydraulic system. Explain pressure control valves in detail with its functions. [8]

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